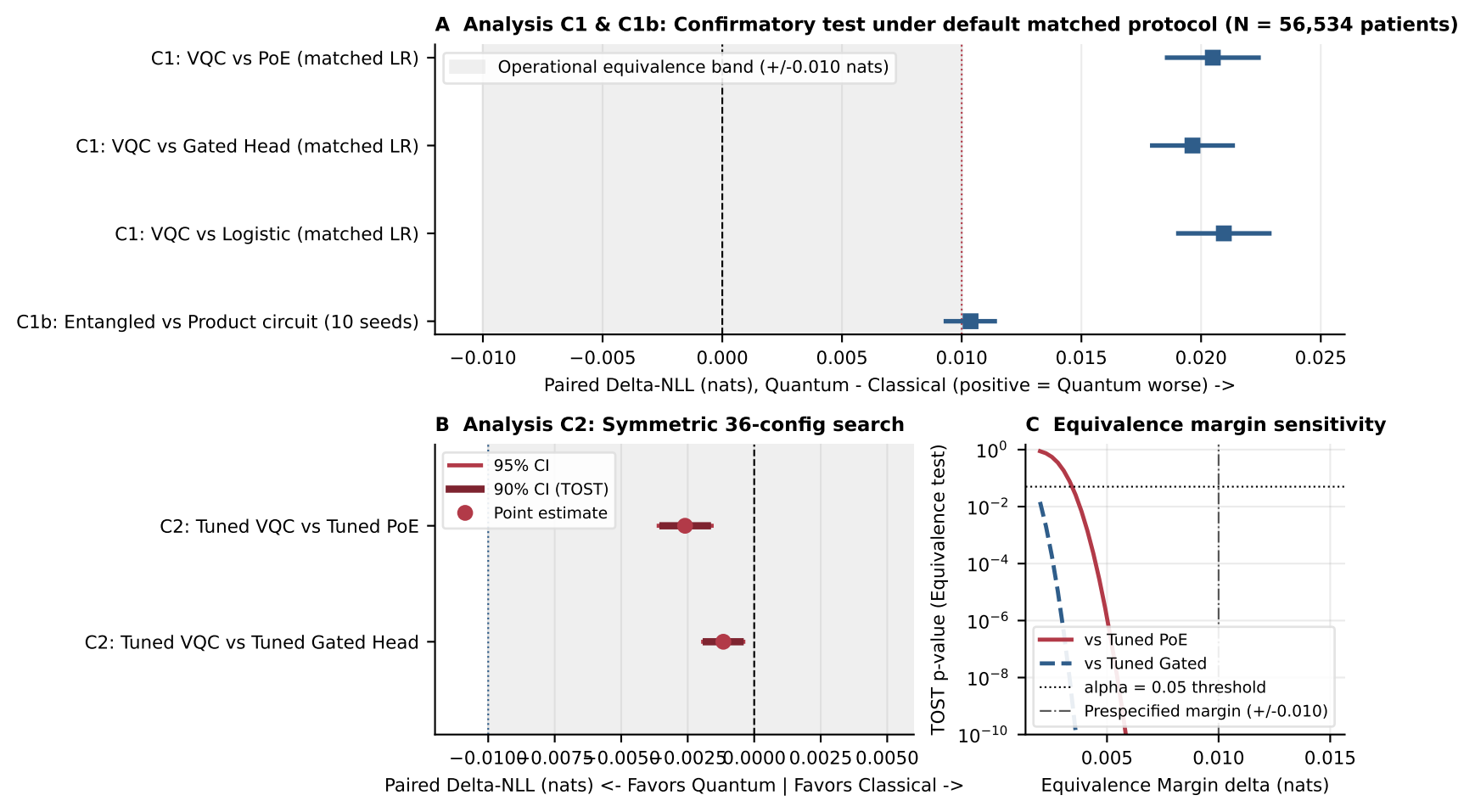
A Analysis C1 & C1b: Confirmatory test under default matched protocol (N = 56,534 patients)
C1: VQC vs PoE (matched LR)
C1: VQC vs Gated Head (matched LR)
C1: VQC vs Logistic (matched LR)
C1b: Entangled vs Product circuit (10 seeds)
Operational equivalence band (+/-0.010 nats)
−0.010
−0.005
0.000
0.005
0.010
0.015
0.020
0.025
Paired Delta-NLL (nats), Quantum - Classical (positive = Quantum worse) ->
B Analysis C2: Symmetric 36-config search
95% CI
90% CI (TOST)
Point estimate
C2: Tuned VQC vs Tuned PoE
C2: Tuned VQC vs Tuned Gated Head
−0.0100
−0.0075
−0.0050
−0.0025
0.0000
0.0025
0.0050
Paired Delta-NLL (nats) <- Favors Quantum | Favors Classical ->
C Equivalence margin sensitivity
TOST p-value (Equivalence test)
10^{0}
10^{-2}
10^{-4}
10^{-6}
10^{-8}
10^{-10}
vs Tuned PoE
vs Tuned Gated
alpha = 0.05 threshold
Prespecified margin (+/-0.010)
0.005
0.010
0.015
Equivalence Margin delta (nats)